DEEPAKKUMAR S

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PROFESSIONAL SUMMARY

Computer Science and Engineering (Data Science) graduate with strong academic foundations in predictive analytics, statistical modeling, and data preprocessing methodologies. Proven track record of applying quantitative research and logical problem-solving to complex datasets during academic coursework. Highly skilled at identifying trends, executing exploratory data analysis, and translating raw data into actionable insights to support business decision-making.

TECHNICAL SKILLS

Languages & Development: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, REST APIs, OOP Concepts
Databases & Backend: MySQL, SQLite, Relational DBMS Concepts, CRUD API Design
AI & Machine Learning: NLP Basics, Scikit-learn, Time-Series Forecasting (ARIMA, SARIMAX, Prophet, LSTM), Generative AI Tools (Gemini, ChatGpt, Claude)
Data Analytics: Data Cleaning, Data Visualization, ETL Concepts
Tools & Practices: Git, GitHub, Tableau, Power BI, Excel, Jupyter Notebook, Agile Basics, Debugging

EDUCATION

B.E. Computer Science and Engineering (Data Science), Annamalai University (2026) - CGPA: 8.72

PROJECT EXPERIENCE

MILO — Offline AI Personal Assistant | Python, Whisper STT, NLP, SQLite

- Cloud-based voice assistants introduce recurring latency bottlenecks and significant privacy vulnerabilities regarding sensitive user audio and data.
- Built an entirely offline, standalone desktop assistant to handle scheduling, reminders, and expense logging without external cloud API dependencies.
- Integrated local OpenAI Whisper STT with custom rule-based NLP intent parsers and structured SQLite persistence for on-device voice execution.
- Achieved 100% on-device data isolation with 0 external network calls, eliminating recurring API costs and cloud data exposure.

Event Management API System | Node.js, TypeScript, Express.js, SQLite, Swagger/OpenAPI

- High-concurrency event registration workflows frequently suffer from duplicate ticketing entries and unstable API contract integrations.
- Architected a maintainable, typed RESTful backend service with validated schema constraints and comprehensive developer documentation.
- Developed modular Express.js micro-routes in TypeScript, enforcing foreign-key constraints in SQLite and generating interactive Swagger/OpenAPI schemas.
- Deployed 15+ fully validated endpoints with zero contract drift, guaranteeing ACID transactional safety across all simultaneous event bookings.

Stock Price Prediction System | Python, ARIMA, SARIMAX, Prophet, LSTM, Scikit-learn, Pandas, NumPy

- Financial equity time-series data exhibits severe non-linear volatility that standard linear regression baselines consistently fail to capture.
- Designed and benchmarked an automated forecasting pipeline to evaluate both classical statistical and deep sequential learning algorithms.
- Built end-to-end data pipelines for sliding-window sequence preprocessing, engineering rolling-window indicators to train ARIMA, Prophet, and LSTM models.
- Achieved an 18% reduction in Root Mean Squared Error (RMSE) over baseline statistical models through fine-tuned LSTM neural networks.

CERTIFICATIONS & WORKSHOPS

- Python & SQL - HackerRank, 2026
- Basics of Python - Infosys Springboard, 2026
- Critical Thinking in the AI Era - HP LIFE, 2026
- Employability Enhancement Training - E-Square (Machine Learning, DBMS, Cloud Concepts, R)
- Event Organizer, COMPSEM Poster Designing Event - Coordinated cross-team logistics